Code No. 3512

OSMANIA UNIVERSITY
FACULTY OF ENGINEERING
UNIVERSITY COLLEGE OF ENGINEERING (AUTONOMOUS)
B.E. (CSE) V-Semester (MAIN) Examinations DEC-2025/ JAN-2026

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Time : 3 hours **Max. Marks : 60**

Note : *i) Answer **Question No. 1** (Compulsory) and answer any **four questions** from the remaining questions (2- 7). Each question caries 12 marks.*
ii) Answers must be written in same order as they occur in the Question Paper.
*iii) **Missing data, if any may suitably be assumed.***

			Marks	BT	CO
1.	a)	List out any four applications of AI in real world.	2	1	2
	b)	Differentiate between perfect and imperfect games.	2	3	2
	c)	What is knowledge acquisition in expert systems?	2	1	3
	d)	Discuss the role of semantic analysis in NLP.	2	2	5
	e)	Explain the role of feedback in learning systems.	2	2	4
	f)	What are the applications that require Fuzzy Logic?	2	1	3
2.	a)	Discuss about generate and test paradigm with an example.	6	3	2
	b)	Distinguish between best first search and Beam search algorithm.	6	3	3
3.	a)	Explain about variations and improvements to minimax.	6	3	3
	b)	Discuss the different types of knowledge representation techniques used in artificial intelligence. Explain each with suitable examples.	6	3	2
4.	a)	Describe fuzzy sets and fuzzy interference with an example.	6	3	3
	b)	Discuss about the characteristics of an expert system.	6	2	3
5.	a)	Explain the ID3 algorithm in detail. Explain how entropy and information gain are used for constructing decision trees.	6	3	4
	b)	Explain the back propagation algorithm. Describe its steps and significance in training multilayer neural networks.	6	3	4

(P.T.O.)

6.	a)	Describe the Chomsky hierarchy of grammar in detail.	6	2	5
	b)	Explain probabilistic models for statistical NLP and their applications in linguistic data analysis.	6	3	5
7.	a)	What is an intelligent agent and explain in detail about types of agents with block diagram for each.	6	3	2
	b)	Explain about alpha beta pruning algorithm with an example.	6	3	2

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